

THE EQUITY EXPRESSION OF THE FOOD SECURITY THESIS

Scarcity With Full Silos

On 30 September the US Department of Agriculture counted 2.1 billion bushels of corn in storage, 35% more than a year earlier, and December corn fell 21 cents to close just above \$5. In the same month the UN's food price index stood at its highest level since 2022, a record El Niño gathered strength in the Pacific, and fertiliser bound for half the world remained stuck behind the Strait of Hormuz. The question is no longer whether food is scarce. It is where the scarcity sits, and who is paying for it.

A reframe worth holding onto

The internal view, carried almost entirely by 13D Research, treats food as the next asset to be repriced as a matter of national security, in the way energy, defence and critical minerals already have been. Its central claim is that a fertiliser shock is worse than an energy shock because crops run on a calendar: a planting window missed for want of nitrogen or phosphate cannot be recovered when the ships start moving again. On this view the harvest that matters is not this one but the next.

Method: external figures (prices, company results, USDA, FAO, NOAA and IFPRI data) are verified and dated in the text and listed in the sources. Internal views come from the Heyokha research Wiki page *Food Security* (last reviewed 25 September 2026) and are attributed to the named research provider. They are opinions, labelled as views, not forecasts or facts.

Picture a bakery whose cellar is full of flour bought last autumn. Across town the mill has lost a third of its grinding stones in a fire, the price of sacks for next year's flour has jumped, and the weather forecaster is warning of a failed wheat crop on another continent. The baker, looking at the cellar, sees no reason to pay more for bread this month, and the bread price barely moves. The miller, who needs sulphur and gas to keep the remaining stones turning, is losing money on every sack. The only question that matters is how long the cellar lasts, and whether the baker can afford next year's sacks before the forecaster turns out to be right. That is the food system in October 2026: a full cellar in the United States, a damaged mill in the Gulf, and a forecast hanging over Asia.

THE STORY IN 60 SECONDS

133.3

FAO Food Price Index in August, up 1.9% on the month, 2.5% on the year and the highest since 2022. Sugar jumped 11.9%, wheat 2.6%. *FAO, released 4 September 2026.*

+35%

US corn stocks at 1 September: 2.095bn bushels, far above trade estimates. December corn fell 21¼ cents to \$5.00¼. *USDA Grain Stocks; Brownfield, 30 September 2026.*

-\$273m

Mosaic's second-quarter net loss, the largest US phosphate maker squeezed by sulphur at \$522 per long ton, rising to \$705 in Q3. *Mosaic results, 4 August 2026.*

WHAT ACTUALLY HAPPENED

Three markets, three different clocks

The food story of 2026 is really three stories running on different timetables. The first is the fertiliser shock. When the Strait of Hormuz closed, roughly a third of globally traded fertiliser lost its route to market. The Gulf states are large exporters of urea and ammonia, both forms of nitrogen, and of sulphur, the raw material for the sulphuric acid without

which phosphate fertiliser cannot be made. The International Food Policy Research Institute reported on 18 September that urea peaked in late April at nearly 80% above its 1 January level. Then it did something the bull case did not anticipate: it fell back to its January baseline by the end of August. Ammonia has followed it most of the way down.

The reason is substitution. Nitrogen is made from natural gas, and there is plenty of gas outside the Gulf. Chinese and American plants ran harder, India bought 52% more nitrogen from other suppliers, and buyers in Europe and Brazil simply bought less. CF Industries ran at 98% of its available ammonia capacity in the first half and reported second-quarter net earnings of \$727m. It estimates the war removed 4.0 to 4.5 million tonnes of traded urea, yet the price has largely absorbed the loss.

Phosphate is a different matter. Sulphur has more than doubled since January and is 40% above where it stood when the Strait closed. China has banned sulphuric acid exports until the end of the year. Phosphate fertiliser prices rose 25–30% and have stayed there. This is the leg of the shock that has not healed, and it shows up in the accounts of the producers themselves: Mosaic, which buys sulphur, lost money in the second quarter even as phosphate prices rose.

The third clock is the grain market, and it is the slowest. The same El Niño that 13D calls the strongest on record has brought rain back to the American corn belt. The USDA's quarterly count showed bins fuller than expected, and corn sold off. Wheat is the exception, up 15% on the year in the FAO's measure after drought cut the American winter crop. The grain price, in other words, is still set by the marginal American bushel, and that bushel is plentiful.

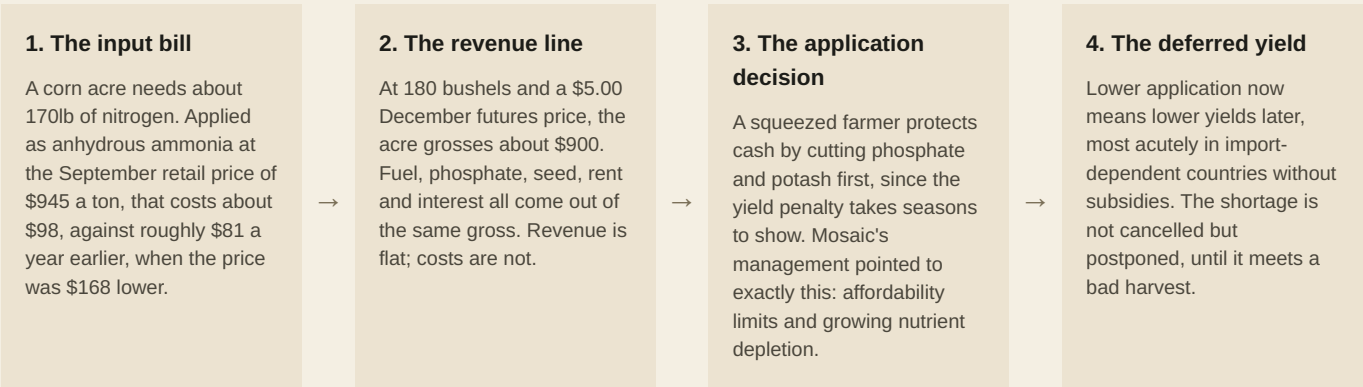
THE ESSENTIAL DISTINCTION

A shortage of food where people live is not the same thing as a shortage of grain where futures are priced. *Bangladesh can run short of fertiliser and Egypt of affordable bread while Chicago corn falls, because the benchmark is set by American storage, and American storage is full.*

THE MECHANISM

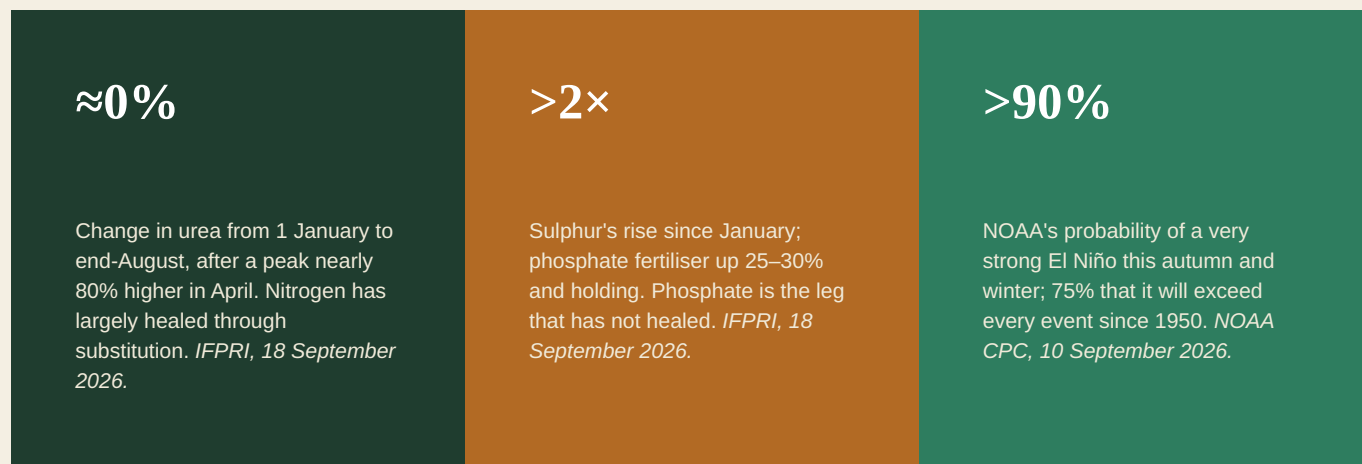
How a fertiliser shock travels to the dinner plate

To see why the internal view insists on patience, follow one acre of American corn through the economics. The arithmetic below is illustrative and uses round numbers; its purpose is to show where the pressure lands, not to model any particular farm.



up only 6.1%. That is a margin squeeze, and margin squeezes change behaviour. What the data cannot yet show is the third and fourth steps, because yields respond to under-fertilisation with a lag measured in seasons.

This is where 13D's view and the market's view part company. 13D argues that food demand is inelastic, so a small supply deficit produces a large price shock, and that many agricultural commodities could rise by 50% to 100%. Goehring & Rozencwajg, whose letter the Wiki also carries, describe a four-year grain bear market that is "living on borrowed time" and expect a multi-year agricultural bull market. But they are careful about timing: they concede that the El Niño has helped the American crop in the short run and that grain may trade sideways for about six months. Fuller American silos are consistent with that caveat.

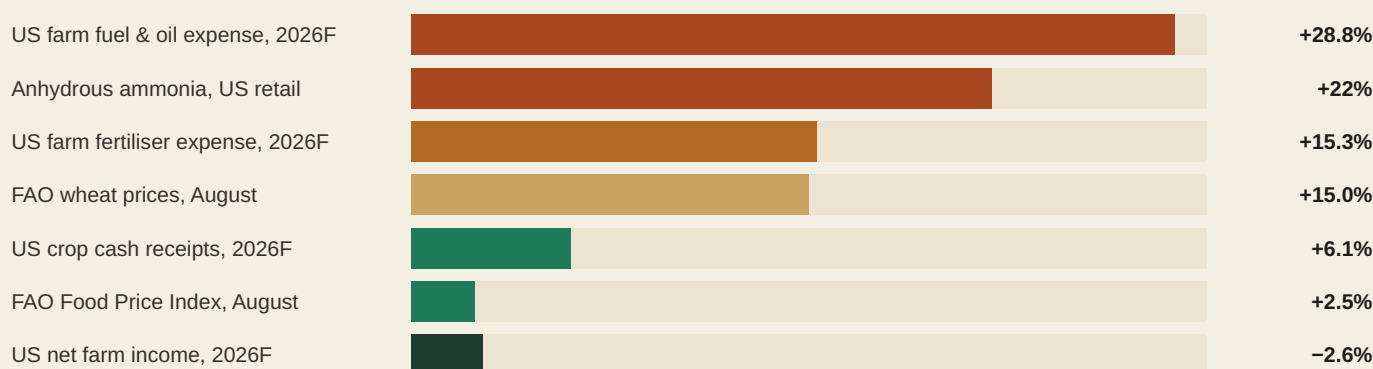


The three numbers the argument turns on: the nitrogen shock has mostly cleared, the phosphate shock has not, and the weather shock has not yet arrived in harvests outside the Americas. Sources as stated in each panel.

THE CENTRAL COMPARISON

Costs are running ahead of receipts

The clearest way to see the squeeze is to set the change in what a crop farmer pays against the change in what food sells for. Every bar below is a year-on-year change.



Bars are illustrative, scaled to the largest value; the net farm income bar represents a decline. Sources: USDA ERS farm income forecast (September 2026, via farmdoc/Farm Policy News); DTN retail fertiliser survey, 14–18 September 2026; FAO Food Price Index, 4 September 2026.

Two things stand out. Input inflation is concentrated in energy-linked costs, fuel and nitrogen, rather than spread evenly. And the food price index, the number the bull case most needs to move, has risen by a fraction of what farmers are paying for inputs. Wheat is the one crop whose price has kept pace. The gap between the top and bottom of the chart is the margin that is being destroyed, and destroyed margins are what eventually reduce supply.

A USEFUL MENTAL MODEL

Treat fertiliser as a loan the soil makes to next year's harvest. *When farmers cannot afford to apply it, nothing happens immediately. The bill comes due one or two seasons later, as lower yields, and arrives at whatever moment the weather happens to choose.*

WHAT HAS ACTUALLY SHIFTED

Six changes, each with two edges

WHAT HAS CHANGED	WHAT IT LOOKS LIKE IN THE DATA	WHY IT CUTS BOTH WAYS
The nitrogen shock has largely passed	Urea back to its 1 January level by end-August after an April peak nearly 80% higher (IFPRI); US retail urea \$659, up 6% year on year (DTN).	Farmers get relief, but the bull case for nitrogen producers loses its price leg. CF's volumes were down 11% in the first half; price did the work.
The phosphate shock has not	Sulphur more than doubled; phosphate up 25–30% and holding; China's acid export ban runs to 31 December.	Reinforces scarcity, yet the phosphate producer most exposed to sulphur, Mosaic, is losing money. Scarcity does not reward every producer.
US grain supply is ample	Corn stocks 2.095bn bushels, up 35%; December corn \$5.00¼ on 30 September.	Undercuts the near-term grain bull case, but low prices deepen the farm margin squeeze that eventually reduces application.
The world price index is rising anyway	FAO index 133.3 in August, highest since 2022; sugar +11.9% in a month; wheat 15% above a year ago.	Suggests stress outside the US benchmark. But a 2.5% annual rise is modest beside 13D's 50–100% scenario.
The El Niño is now record-scale	NOAA: Niño-3.4 at +1.8°C in August; above 90% odds of a very strong event; 75% odds of a historic one.	The worst effects fall on Asia, Australia and Africa. American weather has so far benefited, which keeps the benchmark price subdued.
Policy has entered the potash market	On 21 September the US President said a deal to buy Belarusian potash was being pursued; Nutrien fell 4%, Mosaic more than 5%.	Cheaper potash eases farm costs, the national-security framing in action, but compresses margins for Canadian and US producers.

Both cases, with equal force

The bull case	The bear case
• The weather shock has not yet reached the harvest. NOAA puts the odds of a very strong El Niño above 90%. 13D, citing the World Bank, flags a possible 20–50% loss of rice output in exposed regions, and the Wiki notes the Indian monsoon ran about four weeks late. • Under-fertilisation is a lagged supply cut. Brazil had secured only 30% of its inputs for the 2026/27 soy season by April against a 40% norm (13D, citing industry data); Mosaic reports nutrient depletion risk. Yields respond with a delay. • Phosphate has not normalised. Sulphur more than doubled and China's acid ban runs through December (IFPRI). Mosaic's Q3 sulphur contract settled at \$705 a long ton, up from \$522. • Global prices are rising despite US abundance. The FAO index is at its highest since 2022, with sugar up 11.9% in August alone. • Positioning is light. Goehring & Rozenchwajg argue almost no one is invested in agricultural commodities, with grains 40–50% below their 2022 highs: a contrarian set-up, in their view.	• The benchmark bushel is plentiful. US corn stocks are 35% above a year ago and the price fell below \$5 intraday on the report. Inventory is the market's answer to the scarcity narrative. • Nitrogen healed fast. Urea is back at its January level despite 4.0–4.5mt of lost Gulf urea (CF). Substitution worked faster than the bull case assumed. • Scarcity is not capturing margin for every producer. Mosaic lost \$273m in Q2; CF's volumes fell 11%. Higher prices on lower volume is a fragile basis for earnings. • Policy caps the upside. A Belarus potash deal, food-export bans and subsidies (India's may exceed \$31bn, IFPRI) all transfer value from producers to farmers and consumers. • The farm customer is weakening. USDA sees net farm income down 2.6% and expenses at a record; Deere expects large-equipment sales in the Americas down 15–20% for the rest of its fiscal year.

Sourcing note: the Wiki's own bear section is thin. It lists four general risks, mostly from 2022. The bear case above was therefore built from external reporting (USDA, IFPRI, company results) and from tensions inside the producers' own accounts.

WHERE THE VIEWS DIVERGE

The internal and external evidence conflict most directly on the persistence of the input shock. The Wiki's most recent synthesis carries Goehring & Rozenchwajg's Hormuz-era figures, urea up 70%, as the defining feature of a fertiliser supply shock "pure and simple", and 13D describes Gulf fertiliser plants as physically destroyed rather than rerouted. IFPRI's September data show urea back at its January level and global nitrogen exports down only about 9%. Both can be true: plants may be damaged and the price may still have cleared through substitution and lower demand. But they point to different conclusions about whether the nitrogen producers' windfall is durable. The same tension runs through grain: 13D's 50–100% price scenario sits beside a USDA count showing corn stocks 35% above a year earlier. This edition does not resolve either conflict.

WHAT TO WATCH FROM HERE

The dashboard

SIGNAL	PLAIN-ENGLISH READING	WHY IT MATTERS
FAO Food Price Index, September reading (early October)	Does the world price keep rising while US corn falls?	A further rise would confirm that scarcity is real outside the American benchmark; a fall would weaken the internal view.
USDA WASDE, 9 October	The official US and world balance sheets after the stocks surprise.	Tests whether the ample US supply story extends to world wheat and rice stocks.
Sulphur contract prices for Q4	Whether the input behind phosphate eases or rises again.	The unhealed leg of the shock; it decides whether phosphate makers gain or lose from scarcity.
Southern-hemisphere and Asian crop reports	Planted area and condition in Brazil, Argentina, Australia and India.	Where the El Niño and under-fertilisation should show up first, if they show up at all.

Hormuz fertiliser transits	Whether shipments rise from near zero.	Reopening would remove the national-security premium from producers almost at once.
US potash policy	Whether the Belarus deal is signed and at what price.	Sets the ceiling on North American potash margins for the 2027 season.

THREE POSTURES, NOT ONE ANSWER

Ways to hold the theme, and what each costs

The toll collector

Own the lowest-cost producers of the scarce input: nitrogen makers with cheap American gas, or diversified producers with a retail arm. The logic is that scarcity rents accrue to whoever makes the input. **The cost:** the nitrogen price has already round-tripped, volumes are falling, and a reopening of Hormuz would remove the premium. Policy, as the potash episode showed, can reach in and take the rent.

The deferred harvest

Hold exposure to the crops and regions where the El Niño and under-fertilisation should bite, accepting that the payoff is a matter of seasons, not weeks. **The cost:** time and carry. Goehring & Rozenccwajg themselves expect grain to trade sideways for about six months, and full American silos can hold the benchmark down well beyond any single weather event.

The land underneath

Own farmland, the asset that outlasts any single season and that the Wiki presents as a low-volatility inflation hedge. **The cost:** low income yields, sector multiples that have compressed, and tenants whose margins are being squeezed now. Land is paid by farm profits, and farm profits are falling.

The six companies below are illustrative names chosen to show how the theme reaches listed equities. They are not recommendations, and no view on their valuation or suitability is expressed or implied.

CF

CF Industries Holdings

\$115.20
close, 30 Sep 2026

The purest North American nitrogen producer, running on Henry Hub gas at \$4.01/MMBtu in the first half. Q2 net earnings were \$727m and first-half net earnings \$1.34bn, with ammonia capacity 98% utilised. Volumes fell 11% as price did the work. Management expects nitrogen supply to stay constrained through 2027.

WATCH NEXT

Whether urea's return to January levels shows up in Q3 realised prices.

NTR

Nutrien

\$70.25
close, 30 Sep 2026

The largest potash producer, with nitrogen, phosphate and a farm retail network. Q2 net earnings \$1.22bn, adjusted EBITDA \$2.43bn; potash volume guidance raised to 14.2–14.8mt. First-half nitrogen volumes fell to 4.6mt from 5.5mt. Shares fell 4% on 21 September on the Belarus potash news.

WATCH NEXT

Terms of any US–Belarus potash arrangement and the reaction of North American potash prices.

MOS

The Mosaic Company

\$21.65
close, 30 Sep 2026

The counter-example inside the thesis: a phosphate maker that buys sulphur. Q2 revenue \$2.8bn but a net loss of \$273m; sulphur at \$522 a long ton, with Q3 contracts at \$705. It cut phosphate output and capital spending to \$1.2bn. Realised DAP \$773/t, guided to \$820–840 in Q3.

WATCH NEXT

Whether the DAP price can rise faster than sulphur in Q3 results.

YAR

Yara International (Oslo)

NOK 433.10
close, 25 Sep 2026

Europe's nitrogen champion and a long-standing name in 13D's Food Security Index. Q2 EBITDA excluding special items was \$906m against \$652m a year earlier; net income \$545m. It is buying Gulf Coast Ammonia to shift part of its gas exposure from Europe to the US.

WATCH NEXT

Third-quarter results, and whether European nitrogen demand recovered after the delayed season.

DE**Deere & Company**

\$671.55

close, 30 Sep 2026

A read on the farmer's wallet rather than the input. Q3 FY2026 net income \$1.379bn, but production and precision agriculture sales fell 6% and operating profit 9%. Management expects large-equipment sales in the Americas to fall 15–20% for the rest of the year and calls 2026 the bottom of the cycle.

WATCH NEXT

Fiscal fourth-quarter orders: does the squeeze on farm income deepen the equipment downturn?

FPI**Farmland Partners**

\$10.74

close, 28 Sep 2026

A US farmland REIT named in the Wiki as a scarce real-asset play. It now owns about 70,100 acres in 11 states, far fewer than in 2022. Q2 FFO was 4 cents a share; 2025 revenue fell almost 10%. One broker recently cut its target, citing sector multiple compression.

WATCH NEXT

Q3 results on 28 October, and whether rents hold as crop margins tighten.

The bottom line

What has been repriced so far is the input, not the output. Nitrogen spiked and then cleared; phosphate and sulphur remain dear; farm costs are at a record and farm income is falling. The grain benchmark, set by full American bins, has hardly moved, and the world food price index has risen modestly rather than violently. The equity market has rewarded the nitrogen makers that could run flat out, punished the phosphate maker that buys sulphur, and marked down the farmer's suppliers of machinery.

The evidence is mixed, and it is mixed in a particular way: the external data measure the present, while the internal view is a claim about next season. The reasonable inference is that the margin squeeze on farmers is real and that the supply response it implies has not yet appeared. Whether it appears depends on harvests in Asia, Australia and South America under a record El Niño. The next test is the southern-hemisphere planting data, and the FAO and USDA releases in the first half of October.

Editorial note. This field guide is educational commentary prepared for internal use. It is not investment advice and makes no recommendation to buy, sell or hold any security, commodity or other instrument. Named companies are illustrative. External figures are attributed and dated, and may have changed since. Internal Wiki views are the opinions of the named research providers, labelled as views; they are not verified facts or forecasts. Where internal and external views conflict, the conflict is stated, not resolved.

Sources and update notes

Internal. Heyokha Wiki theme page *Food Security* (created 23 June 2026; last reviewed 25 September 2026), sections Thesis, The bull case, The bear case & what breaks it, Evidence & data, Contradictions, and the dated 2026 update sections (Q1 2026; April 2026; August 2026; September 2026). Provider issues relied on: 13D Research WILTW 2 April 2026, 30 April 2026, 27 August 2026, 10 September 2026 ("Diesel is the transmission mechanism") and 24 September 2026; 13D WATMTU 16 August 2026 and 20 September 2026; Goehring & Rozencwajg natural resource commentaries of 19 May 2026 (Q1 2026) and 28 August 2026; Greed & Fear (Christopher Wood) 16 July 2026; CLSA Bits & Pieces 11 September 2026 (Jeff Currie); Gary Shilling's Insight, 1 August 2026.

Limitation on the internal view. The Food Security page is heavily concentrated in a single provider. Of the 32 source issues listed in its frontmatter, 30 are 13D Research publications, many dating from 2022–2024; the remaining two are Goehring & Rozencwajg and Greed & Fear. 13D also runs a Food Security Index and a portfolio allocation to the theme, so its analysis is not disinterested. The page's bear section is brief and largely dated to 2022, and its Contradictions section notes no internal disagreement. The internal view should therefore be read as one well-argued house view, not a consensus.

External sources. (1) FAO, Food Price Index release, 4 September 2026: August index 133.3, +1.9% m/m, +2.5% y/y; cereals +2.2%, wheat +2.6% m/m and 15.0% above a year earlier, sugar +11.9%, dairy +2.3%, vegetable oils +1.1%, meat +1.0%. (2) USDA NASS Grain Stocks, 30 September 2026, via Brownfield Ag News: 1 September corn stocks 2.095bn bu (+35% y/y), wheat 1.846bn bu (–14%), soybeans 315.1m bu. (3) Brownfield, closing futures 30 September 2026: December corn \$5.00¾ (–21¼c), November soybeans \$12.93 (–4¾c), December Chicago wheat \$6.75¾ (–17c). (4) IFPRI blog, 18 September 2026: urea peaked nearly 80% above 1 January in late April and returned to that baseline by end-August; ammonia near pre-crisis levels; MAP/DAP +25–30% and holding; sulphur more than doubled year to date and +40% since closure; global N and P exports each about –9% vs 2025; India N imports +52%, EU –51%, Brazil –19%; India subsidies may exceed \$31bn. (5) DTN retail fertiliser survey, 14–18 September 2026: anhydrous \$945/t (+22% y/y),

urea \$659 (+6%), DAP \$925 (+2%), MAP \$967 (+5%), potash \$495 (+2%), UAN28 \$421, UAN32 \$458 (−3%). (6) USDA ERS farm income forecast, September 2026, via Farm Policy News (University of Illinois): net farm income \$158.4bn (−2.6%; −5.5% inflation-adjusted), expenses \$492.8bn (+\$21.2bn), fertiliser +\$5.3bn (+15.3%), fuel and oil +\$4.8bn (+28.8%), crop receipts \$253bn (+6.1%). (7) NOAA CPC ENSO Diagnostic Discussion, 10 September 2026: Niño-3.4 +1.8°C in August; above 90% chance of a very strong event; 75% chance of exceeding events since 1950 in October–December. (8) Mosaic Q2 2026 release, 4 August 2026: revenue \$2.8bn, net loss \$273m, adjusted EBITDA \$407m; sulphur \$522/long ton, Q3 contracts \$705; ammonia \$621/t; DAP \$773/t realised, Q3 guide \$820–840. (9) CF Industries H1 2026 release, August 2026: Q2 net earnings \$727m; H1 net earnings \$1.34bn, adjusted EBITDA \$2.18bn; volumes −11% y/y; gas \$4.01/MMBtu; Middle East traded urea supply reduced by 4.0–4.5mt. (10) Nutrien Q2 2026 release, 5 August 2026: net earnings \$1.22bn, adjusted EBITDA \$2.43bn, adjusted EPS \$2.61; H1 nitrogen volumes 4.594mt vs 5.486mt. (11) Yara Q2 2026 release, 17 July 2026: EBITDA ex special items \$906m vs \$652m; net income \$545m. (12) Deere Q3 FY2026 release, 20 August 2026: net income \$1.379bn; production & precision ag sales −6%, operating profit −9%; FY net income guide \$4.75–5.00bn. (13) Investing.com, 21 September 2026: on the Belarus potash remarks NTR −4%, MOS −5% +, CF −3%. (14) Share prices: stockanalysis.com (CF, NTR, MOS, DE closes 30 September 2026; FPI close 28 September 2026); ad-hoc-news.de (Yara close NOK 433.10, 25 September 2026). (15) USDA WASDE date of 9 October 2026 per the agsist.com USDA calendar.

Editorial choices made in this automated run. Food Security was selected by the house rule: themes were scored on their appearance in Wiki sources published in the last 21 days (none fell in the last three days; the newest issues are dated 24 September). It scored 7, appearing in five of the thirteen recent issues, including 13D's 10 and 24 September issues and CLSA's 11 September report. Higher-scoring themes were skipped on cooldown, each having its own edition in the past 30 days: Gold (28 September), AI Capex (8 September), Critical Minerals (21 September), Oil & Gas (21 September) and Copper (24 September). The archive folder was read through the device bridge and all filenames were parsed; the 31 August "Grain Supply Shock" edition is closely related but falls just outside the 30-day window, and this edition deliberately centres on the fertiliser-margin mechanism rather than grain supply. External prices were obtained by web search and fetch only; yfinance and direct market-data access are not available in this environment. Yara's price is the latest dated Oslo close found (25 September). No tool failures affected the content.

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